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Whole-body Humanoid Robot Locomotion with Human Reference

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***Abstract*— Recently, humanoid robots have made significant advances in their ability to perform challenging tasks due to the deployment of Reinforcement Learning (RL), however, the inherent complexity of humanoid robots, including the difficulty of designing complicated reward functions and training entire sophisticated systems, still poses a notable challenge. To conquer these challenges, after many iterations and in-depth investigations, we have meticulously developed a full-size humanoid robot, "Adam", whose innovative structural design greatly improves the efficiency and effectiveness of the imitation learning process. In addition, we have developed a novel imitation learning framework based on an adversarial motion prior, which applies not only to Adam but also to humanoid robots in general. Using the framework, Adam can exhibit unprecedented human-like characteristics in locomotion tasks. Our experimental results demonstrate that the proposed framework enables Adam to achieve human-comparable performance in complex locomotion tasks, marking the first time that human locomotion data has been used for imitation learning in a full-size humanoid robot. For more video demonstrations, please visit our YouTube channel: https://www.youtube.com/watch?v=7hK2ySYBa1I**



Fig. 1: The top image displays the humanoid robot Adam walking on unseen terrain, and the bottom image shows Adam moving from a standing position to running.

I. Introduction

In recent years, the field of humanoid robotics has received widespread attention, with numerous research institutes and companies releasing cutting-edge innovations and research results successively, signifying the rapid development and rise of the field. Boston Dynamics' Atlas robot [1] has demonstrated parkour-level mobility; Tesla's Optimus [1] and Figure's humanoid robots [2] have learned from human data to perform complex desktop manipulation tasks; the bipedal robot Cassie [2] [3] [4] and its humanoid version Digit [3], are powered by motors and successfully move across a variety of terrains; the renowned legged robot company Unitree released their humanoid robot product H1 [4]; Apptronik [5] has developed a humanoid robot named Apollo, powered entirely by push rod electric motors; OpenAI [6], esteemed in the field of general-purpose AI, has acquired the 1X robot company [7] and proposed a development plan for embodied intelligence. The above indicates that humanoid robotics is becoming one of the key directions for researchers and companies and that mastering the core technology of humanoid robotics is crucial to bridging the gap between digital general-purpose AI and tangible hardware.

Traditional robot control algorithms, which typically rely on precise mathematical models and predefined motion planning, have proven to be very effective in the past for locomotion tasks of quadrupedal, bipedal, and humanoid robots. Notably, Boston Dynamics' Atlas and Spot robots have demonstrated the effectiveness of these methods by using Model Predictive Control (MPC) algorithms [5] to present impressive extreme mobility in a variety of demonstrations. However, these algorithms often rely on accurate modeling of the environment, which can introduce significant challenges in terms of robustness and generalisability, especially in unknown or dynamically changing environments, where the performance of traditional control algorithms can be significantly degraded, limiting their usefulness in a wider range of application scenarios. This reliance on accurate modeling, moreover, requires a high level of expertise to build and maintain these models, increasing the complexity of development and debugging.

Traditional robot control algorithms are notably limited in their adaptability, flexibility, and user-friendliness even though they exhibit excellent performance in specific environments, motivating researchers to explore alternative approaches to overcome these obstacles to design smarter and more adaptable robot control strategies. Among them, deep neural network-based reinforcement learning algorithms have

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基于人类运动参考的全身人形机器人运动

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摘要：近年来，随着强化学习（***RL***）的应用，人形机器人在执行挑战性任务方面的能力已取得显著进步；然而，人形机器人固有的复杂性，包括设计复杂奖励函数以及训练整套精密系统的难度，仍然构成显著挑战。为了克服这些挑战，经过多次迭代与深入研究，我们精心研发了全尺寸人形机器人***"Adam"***，其创新的结构设计大幅提升了模仿学习过程的效率与效果。此外，我们提出了一种基于对抗式运动先验的新型模仿学习框架，该框架不仅适用于***Adam***，也普遍适用于人形机器人。借助该框架，***Adam***能够在运动任务中展现出前所未有的类人特征。实验结果表明，该框架使***Adam***在复杂运动任务中达到与人类相当的表现，这标志着人类运动数据首次被用于全尺寸人形机器人的模仿学习。更多视频演示，请访问我们的***YouTube***频道：***https://www.youtube.com/watch?v=7hK2ySYBa1I***



图 1：上图展示了人形机器人Adam在陌生地形上行走，下图展示了Adam从站立状态到跑步状态的过渡。

I. 引言

近年来，人形机器人领域受到广泛关注，众多研究机构和公司相继发布前沿创新与研究成果，标志着该领域的快速发展和崛起。波士顿动力公司的Atlas机器人[1]展示了跑酷级别的运动能力；特斯拉的Optimus [1]和Figure公司的人形机器人[2]已通过人类数据学会了执行复杂的桌面操作任务；双足机器人Cassie [2] [3] [4]及其人形版本Digit 3均由电机驱动，并能够成功穿越多种地形；知名足式机器人公司宇树科技发布了其人形机器人产品H1 4；Apptronik [5]开发了一款名为Apollo的人形机器人，它完全由推杆电机驱动；在通用人工智能领域备受推崇的OpenAI 6收购了1X机器人公司[7]，并提出了具身智能的发展规划。上述情况表明，人形机器人正成为研究人员和企业关注的重点方向之一，也表明掌握人形机器人核心技术是对于弥合数字通用人工智能与实体硬件之间的差距至关重要。

传统机器人控制算法通常依赖精确的数学模型和预定义的运动规划，过去已被证明在四足、双足和人形机器人的运动任务中非常有效。值得注意的是，波士顿动力公司的Atlas和Spot机器人通过使用模型预测控制（MPC）算法 [5] 在各种演示中展现了令人印象深刻的极限机动性，验证了这些方法的有效性。然而，这些算法通常依赖环境的精确建模，这在鲁棒性和泛化能力方面会带来重大挑战，尤其是在未知或动态变化的环境中，传统控制算法的性能可能会显著下降，从而限制了它们在更广泛应用场景中的实用性。此外，这种对精确建模的依赖需要高水平的专业知识来构建和维护这些模型，增加了开发和调试的复杂性。

尽管传统机器人控制算法在特定环境中表现出优异的性能，但其适应性、灵活性和易用性明显受限，这促使研究人员探索替代方法，以克服这些障碍，从而设计出更智能、更具适应性的机器人控制策略。其中，基于深度神经网络的强化学习算法已经

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achieved promising results in the control of legged robots [6]. By interacting with the environment, Deep reinforcement learning algorithms are able to autonomously discover effective strategies to perform complex tasks and can be potentially extended to unknown or dynamically changing environments, which provides robots with unprecedented adaptability and flexibility.

Reinforcement learning algorithms have made significant progress in a variety of legged robots, but applications in the field of humanoid robotics still lack sufficient exploration. This can be mainly attributed to the following factors: Firstly, most humanoid robots are expensive and difficult to maintain, which is a considerable obstacle for research institutes with limited funds; secondly, the interpretability problem of deep neural networks and the Sim2Real gap in the training process of deep reinforcement learning make it difficult to transfer the models to practical applications; Lastly, the complexity of humanoid robots greatly exceeds that of other legged robots, which makes the design of reward functions and training strategies during their training more challenging.

To address these challenges, after repeated design and in-depth exploration, we have introduced the motor-joint-driven humanoid robot "Adam", which has a significant cost advantage over traditional hydraulically-driven robots, and its modular design facilitates repairs during experiments and further reduces maintenance costs. Moreover, our high-performance actuators ensure the robot's exceptional mobility, granting it a range of motion in its limbs close to that of a human. For the difficulty of setting up complicated reward functions in reinforcement learning, we adopt an innovative strategy of using human motion data to guide the learning process. Combined with our imitation learning training framework, Adam's performance on the locomotion task in the experiment is impressive.

In summary, we introduce a brand-new humanoid robot, Adam, and provide a new methodology and experimental validation for the learning, adaptation, and optimization of humanoid robots, paving a new way for research and development in humanoid robotics. The contributions of this paper can be summarized in the following three points:

1) We have developed and detailed an innovative biomimetic humanoid robot, Adam, whose limbs not only have a range of motion close to that of humans but also offer sufficient advantages in terms of low cost and ease of maintenance.
2) We designed and validated a new whole-body imitation learning framework for humanoid robots, which effectively solves the problem of complex reward function settings encountered in the reinforcement learning training of humanoid robots, greatly reduces the Sim2Real gap and improves the learning ability and adaptability of humanoid robots.
3) To address the Sim2Real challenges in complex humanoid robot reinforcement learning control algorithms, we incorporated numerous cross-validation and feedback adjustment steps into our framework. We not only demonstrated the robot's highly human-like performance in executing complex motion tasks but also provided a new perspective and data support for the future motion learning and optimization of humanoid robots.

II. Related work

A. Legged robots locomotion

Legged robot locomotion has seen significant advancement as it serves as the fundamental basis for achieving humanoid robotics. The incorporation [7], [8] of Reinforcement Learning (RL) has played a crucial role in this field. The robot Cassie realizes a wide range of walking and running patterns utilizing periodic-parametrized reward functions [9], and even set a Guinness World Record for the fastest 100m dash using pre-optimized gaits [10]. Jeon et al. [11] underscores the effectiveness of potential-based reward shaping in expediting learning and enhancing the robustness of legged locomotion. Similarly, Shi et al. [12] have broadened the capabilities of legged robots by introducing an assistive force curriculum that facilitates agile motion learning in settings without explicit references. The HRP-5P humanoid robot has demonstrated exceptional bipedal walking by leveraging feedback from actuator currents [13], while Kim et al. [14] have proposed a torque-based approach to bridge the gap between simulated training and real-world application effectively. In an innovative endeavor, DeepMind's research [15] enabled a miniature humanoid robot to master complex soccer skills through a unique teacher-student distillation and self-play methodology. Additionally, utilizing attention-based transformers in the Digit humanoid robot has facilitated more adaptable and versatile locomotion patterns [16]. More recently, Tang et al. [17] employed adversarial critic component and specially designed Wasserstein distances to migrate locomotion from human reference. However, Adam has more flexibility and is more human-like overall, which leads to better performance in real robot experiments.

B. Learning from Human Reference

Humans, with their advanced intelligence and versatile locomotion capabilities, exhibit complex motion patterns that embody rich information. Leveraging insights from human behavior through learning from human references can greatly enhance the adaptability of robots. Traditional approaches to behavior cloning relied on manual programming, which proved to be time-consuming and inflexible [18], [19]. Furthermore, defining the intricate and versatile locomotion skills of humanoid robots manually posed significant challenges [20], [21].

In recent years, imitation learning (IL) strategies have gained prominence, involving the tracking of either reference joint trajectories or extracted gait features [22]–[25]. However, these tracking techniques often operate on individual motion clips, resulting in discontinuities when transitioning between different locomotion patterns. To address this limitation, Generative Adversarial Imitation Learning (GAIL) was introduced by Peng et al., who proposed two innovative methods: AMP and Successor ASE [26]–[28].

在腿式机器人 [6]的控制方面取得了令人瞩目的成果。通过与环境的交互，深度强化学习算法能够自主发现执行复杂任务的有效策略，并有可能扩展到未知或动态变化的环境中，这为机器人提供了前所未有的适应性和灵活性。

强化学习算法已在多种足式机器人上取得了显著进展，但在人形机器人领域的应用仍缺乏充分探索。这主要可归因于以下因素：首先，大多数人形机器人价格昂贵且难以维护，这对资金有限的研究机构而言是一个相当大的障碍；其次，深度神经网络的可解释性问题以及深度强化学习训练过程中的Sim2Real（仿真到现实）差距，使得模型难以迁移到实际应用中；最后，人形机器人的复杂性远超其他足式机器人，这使其训练过程中的奖励函数设计和训练策略制定更具挑战性。

为了应对这些挑战，经过反复设计与深入探索，我们推出了电机关节驱动的人形机器人"Adam"。与传统液压驱动机器人相比，它具有显著的成本优势，其模块化设计便于在实验中进行维修，并进一步降低了维护成本。此外，我们高性能的执行器确保了机器人出色的运动能力，使其四肢的活动范围接近人类。针对强化学习中复杂奖励函数设置困难的问题，我们采用了一种创新策略，即利用人体运动数据来引导学习过程。结合我们的模仿学习训练框架，Adam在实验中的运动任务表现令人印象深刻。

总之，我们介绍了一款全新的人形机器人 Adam，并为人形机器人的学习、适应和优化提供了新的方法和实验验证，为人形机器人领域的研究与开发开辟了一条新道路。本文的贡献可概括为以下三点：

1) 我们开发并详细介绍了一款创新的仿生人形机器人 Adam，它的四肢不仅具有接近人类的运动范围，而且在低成本和易维护方面也具备显著优势。

2) 我们设计并验证了一种面向人形机器人的全新全身模仿学习框架，有效解决了人形机器人强化学习训练中奖励函数设置复杂的问题，大幅缩小了 Sim2Real 差距，并提升了人形机器人的学习能力和适应能力。

3) 为了解决复杂人形机器人强化学习控制算法中的 Sim2Real 挑战，我们在框架中融入了大量交叉验证和反馈调整步骤。我们不仅展示了该机器人高度类人的在执行复杂运动任务时的表现，同时也为未来人形机器人的运动学习与优化提供了新的视角和数据支持。

II. 相关工作

A. 腿足式机器人运动

腿足式机器人运动作为实现人形机器人的基础，已经取得了显著进展。强化学习（RL）的引入 [7], [8]在这一领域发挥了关键作用。Cassie 机器人利用周期参数化奖励函数 [9],实现了多种行走和奔跑模式，甚至使用预先优化的步态 [10]创造了100米短跑最快的吉尼斯世界纪录。Jeon 等人 [11] 强调了基于势函数的奖励塑形在加速学习与增强腿足式运动鲁棒性方面的有效性。类似地，Shi 等人 [12] 通过引入辅助力课程，扩展了腿足式机器人的能力，使其在无显式参考的情况下也能进行敏捷运动学习。HRP-5P 人形机器人借助执行器电流反馈 [13], 展现了出色的双足行走能力，同时 Kim 等人 [14] 提出了一种基于扭矩的方法，有效弥合了仿真训练与实际应用之间的差距。在一项创新尝试中，DeepMind 的研究 [15]通过独特的师生蒸馏与自博弈方法，使一款微型人形机器人掌握了复杂的足球技能。此外，Digit 人形机器人利用基于注意力的 Transformer，也促进了更具适应性和多样性的运动模式 [16]。最近，Tang 等人 [17] 采用对抗性评论器组件和专门设计的 Wasserstein 距离，将运动技能从人类参考动作中迁移过来。然而，Adam 整体上更加灵活、更像人类，这使其在真实机器人实验中表现更佳。

B. 从人类参考中学习

人类凭借其先进的智能和多样化的运动能力，展现出蕴含丰富信息的复杂运动模式。通过从人类参考中学习，借鉴人类行为的洞见，可以极大增强机器人的适应性。传统的行为克隆方法依赖手动编程，而事实证明这种方法既耗时又缺乏灵活性 [18], [19]。此外，手动定义人形机器人复杂且多样化的运动技能也构成了重大挑战 [20], [21]。

近年来，模仿学习（IL）策略日益受到重视，其涉及对参考关节轨迹或提取的步态特征进行跟踪 [22]–[25]。然而，这些跟踪技术通常作用于单个运动片段，导致在不同运动模式之间切换时出现不连续。为解决这一局限性，Peng等人引入了生成对抗模仿学习（GAIL），并提出了两种创新方法：AMP和Successor ASE [26]–[28]。

These approaches enable physics-based avatars to perform objective tasks while implicitly imitating diverse motion styles from extensive unstructured datasets. Variants of AMP have been successfully applied to learn agile quadrupedal locomotion and terrain-adaptive skills [29]–[33]. Additionally, Tang et al. introduced a Wasserstein adversarial imitation system with soft boundary constraints, further enhancing the capabilities of the AMP method [17]. To facilitate the transfer of reference motion to robots, many works have introduced re-targeting techniques [17], [34], [35] that consider primitive skeleton and geometry consistency, enabling accurate dynamic modeling and complex balance controllers. However, AMP only makes robots learn kinematic motion relationships, which lacks physics constraints and causes a huge sim-to-real gap.

III. Preliminary

A. The Structure of Humanoid Robot Adam

In this paper, we conducted experiments using the lite version of Adam. Adam(Lite) is equipped with 25 QDD(quasi-direct drive), force-controlled PND actuators throughout its body, standing at 1.6 meters tall and weighing 60 kilograms. Its legs are fitted with four QDD high-sensitivity, highly back-drivable actuators with up to 340N·m maximum torque. The arms have five degrees of freedom, and the waist has three. This fully modular, highly reusable design of flexible actuators, along with a highly biomimetic torso configuration, provides Adam with exceptional mobility and adaptability. The entire body utilizes a full-stack in-house design, real-time communication network PDN (PND Network), and PND actuators. The motion control computer is the 12th generation Intel i7 processor (Intel NUC) and PND RCU(Robot control unit). The PND RCU integrates all actuators, battery management system (BMS), power management, and is equipped with a 16-port Gigabit Ethernet programmable switch with network management capabilities, forming the robot's sensory and control communication hub. This configuration enables Adam to perform large-scale parallel dynamic simulations and neural network training, achieving diversified full-body motion control that is truly applicable to real service scenarios, adapting to the complex and variable human social environment. Dexterous hands and vision modules can be optionally equipped. Since we focused on blind locomotion tasks in this paper, these parts were not included. The schematic diagram of Adam(Lite)'s entire body structure is shown in Figure. 2, with information on its joint structure and more detailed in TABLE. I.

TABLE I: Specifications of Humanoid Robot Adam.

Humanoid Robot	Adam (Lite)
Height	1.6 M
Weight	60 Kg
Full-body degrees of freedom	25
Single-leg degrees of freedom	Hip x3 + Knee x1 + Ankle x2
Single-arm degrees of freedom	Shoulder x3 + Elbow x2
Waist degrees of freedom	3

B. Motion Capture and Re-Targeting

In our research, we explored a variety of human motion data sources to enrich our training set, ensuring that our model could learn diverse human motion characteristics. Initially, we utilized two public motion capture (mocap) databases: the SFU mocap dataset [8] and the CMU mocap dataset [9]. Both datasets contain multiple motion capture sequences, covering a wide range of human activities, including everyday actions, sports movements, dance, and combat actions. By integrating these two databases, we provided Adam with a diverse and high-quality human whole-body motion dataset, which is crucial for training the robot to understand and mimic human motion patterns.

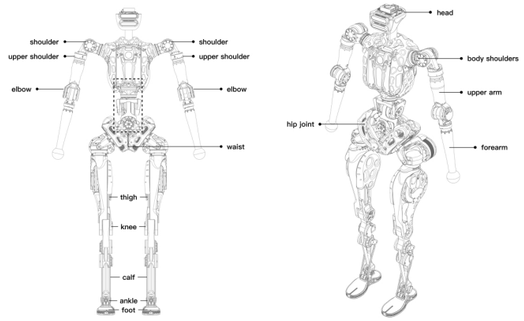


Fig. 2: Schematic representation of the humanoid robot Adam(Lite), showing both the front view and the angled side view. Particularly, the design of its hip joint distinctly emulates the human skeleton, demonstrating a remarkable level of anthropomorphism throughout its structure.

Beyond public datasets, we also used high-precision motion capture equipment for custom motion recording. This approach allowed us to capture specific motion data, especially those special actions or sequences designed for particular experimental needs that are difficult to find in public databases. These custom motion data not only added diversity to our dataset but also enabled us to fine-tune and optimize our model more precisely, adapting it to specific motion tasks and challenges. We did not approach re-targeting as an optimization problem, as was done in approaches like humanmimic [17]. Our objective was to secure high-quality motion data tailored for Adam, which led us to manually calibrate and convert each dataset ourselves.

C. Reinforcement Learning on Legged Robots

The application of RL to legged robots is modeled as a Partially Observable Markov Decision Process (POMDP), defined by the tuple $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p, \gamma)$ where $\mathcal{S}$ defines the state space, including all states of the environment. $\mathcal{O}$ denotes the partial observation space, representing a subset of the state space $\mathcal{S}$ that covers only the observable aspects of the environment by the agent. Concurrently, $\mathcal{A}$ specifies the action space, indicating all actions the agent is capable of executing. The reward function, denoted by $\mathcal{R}$, assigns a scalar reward to each state-action pair. The transition probabilities, represented by p, specify the likelihood of transition from the current state to a new state following the chosen action. The discount factor $\gamma \in [0, 1]$, is applied to ensure that future rewards are appropriately weighted.

Within this framework, at each timestep t, the agent observes $o_t \in \mathcal{O}$ from the environment. Based on this ob-

These approaches enable physics-based avatars to perform objective tasks while implicitly imitating diverse motion styles from extensive unstructured datasets. Variants of AMP have been successfully applied to learn agile quadrupedal locomotion and terrain-adaptive skills [29]–[33]. Additionally, Tang et al. introduced a Wasserstein adversarial imitation system with soft boundary constraints, further enhancing the capabilities of the AMP method [17]. To facilitate the transfer of reference motion to robots, many works have introduced re-targeting techniques [17], [34], [35] that consider primitive skeleton and geometry consistency, enabling accurate dynamic modeling and complex balance controllers. However, AMP only makes robots learn kinematic motion relationships, which lacks physics constraints and causes a huge sim-to-real gap.

III. Preliminary

A. The Structure of Humanoid Robot Adam

In this paper, we conducted experiments using the lite version of Adam. Adam(Lite) is equipped with 25 QDD(quasi-direct drive), force-controlled PND actuators throughout its body, standing at 1.6 meters tall and weighing 60 kilograms. Its legs are fitted with four QDD high-sensitivity, highly back-drivable actuators with up to 340N·m maximum torque. The arms have five degrees of freedom, and the waist has three. This fully modular, highly reusable design of flexible actuators, along with a highly biomimetic torso configuration, provides Adam with exceptional mobility and adaptability. The entire body utilizes a full-stack in-house design, real-time communication network PDN (PND Network), and PND actuators. The motion control computer is the 12th generation Intel i7 processor (Intel NUC) and PND RCU(Robot control unit). The PND RCU integrates all actuators, battery management system (BMS), power management, and is equipped with a 16-port Gigabit Ethernet programmable switch with network management capabilities, forming the robot's sensory and control communication hub. This configuration enables Adam to perform large-scale parallel dynamic simulations and neural network training, achieving diversified full-body motion control that is truly applicable to real service scenarios, adapting to the complex and variable human social environment. Dexterous hands and vision modules can be optionally equipped. Since we focused on blind locomotion tasks in this paper, these parts were not included. The schematic diagram of Adam(Lite)'s entire body structure is shown in Figure. 2, with information on its joint structure and more detailed in TABLE. I.

TABLE I: Specifications of Humanoid Robot Adam.

Humanoid Robot	Adam (Lite)
Height	1.6 M
Weight	60 Kg
Full-body degrees of freedom	25
Single-leg degrees of freedom	Hip x3 + Knee x1 + Ankle x2
Single-arm degrees of freedom	Shoulder x3 + Elbow x2
Waist degrees of freedom	3

B. Motion Capture and Re-Targeting

In our research, we explored a variety of human motion data sources to enrich our training set, ensuring that our model could learn diverse human motion characteristics. Initially, we utilized two public motion capture (mocap) databases: the SFU mocap dataset [8] and the CMU mocap dataset [9]. Both datasets contain multiple motion capture sequences, covering a wide range of human activities, including everyday actions, sports movements, dance, and combat actions. By integrating these two databases, we provided Adam with a diverse and high-quality human whole-body motion dataset, which is crucial for training the robot to understand and mimic human motion patterns.

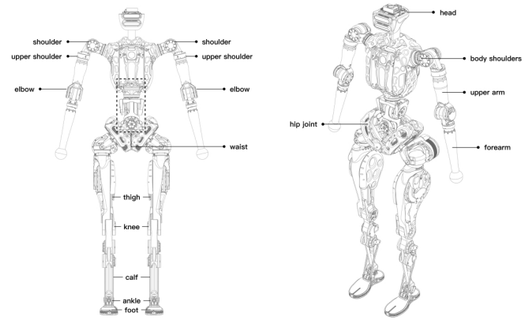


Fig. 2: Schematic representation of the humanoid robot Adam(Lite), showing both the front view and the angled side view. Particularly, the design of its hip joint distinctly emulates the human skeleton, demonstrating a remarkable level of anthropomorphism throughout its structure.

Beyond public datasets, we also used high-precision motion capture equipment for custom motion recording. This approach allowed us to capture specific motion data, especially those special actions or sequences designed for particular experimental needs that are difficult to find in public databases. These custom motion data not only added diversity to our dataset but also enabled us to fine-tune and optimize our model more precisely, adapting it to specific motion tasks and challenges. We did not approach re-targeting as an optimization problem, as was done in approaches like humanmimic [17]. Our objective was to secure high-quality motion data tailored for Adam, which led us to manually calibrate and convert each dataset ourselves.

C. Reinforcement Learning on Legged Robots

The application of RL to legged robots is modeled as a Partially Observable Markov Decision Process (POMDP), defined by the tuple $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p, \gamma)$ where $\mathcal{S}$ defines the state space, including all states of the environment. $\mathcal{O}$ denotes the partial observation space, representing a subset of the state space $\mathcal{S}$ that covers only the observable aspects of the environment by the agent. Concurrently, $\mathcal{A}$ specifies the action space, indicating all actions the agent is capable of executing. The reward function, denoted by $\mathcal{R}$, assigns a scalar reward to each state-action pair. The transition probabilities, represented by p, specify the likelihood of transition from the current state to a new state following the chosen action. The discount factor $\gamma \in [0, 1]$, is applied to ensure that future rewards are appropriately weighted.

Within this framework, at each timestep t, the agent observes $o_t \in \mathcal{O}$ from the environment. Based on this ob-

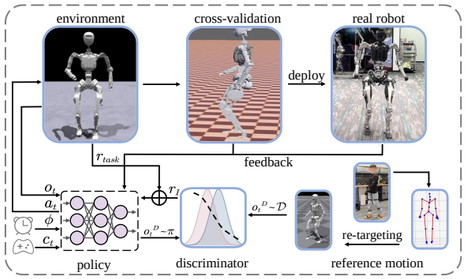


Fig. 3: Adversarial Motion Priors Imitation Training Framework of Humanoid Robot

servation, the agent outputs an action $a_t \in \mathcal{A}$ sampled from a policy $\pi(a_t|o_t)$. Subsequently, the environment transitions to a new state s_{t+1}, governed by the probability distribution $s_{t+1} \sim p(s_{t+1}|s_t, a_t)$, and the agent is accordingly awarded a reward $r_t = \mathcal{R}(s_t, a_t)$. This objective function is formalized as to maximize the received reward:

$$\arg\max_{\theta} \mathbb{E}_{(s_t,a_t)\sim p_\theta(s_t,a_t)} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right] \tag{1}$$

where T denotes the time horizon of POMDP.

IV. Method

A. Learning with Human Reference by Adversarial Motion Priors

Our humanoid imitation framework is built upon the AMP. In our framework, the discriminator D outputs the similarity between the state transitions sampled from the agent and it sampled from the reference demonstrations $\mathcal{D}$. It is crucial to select the discriminator observation $o_t^D \in \mathbb{R}^{58}$ fed into the discriminator to ensure the robots with similar state transitions can perform similar locomotion styles. The discriminator observation contains the velocity, position of each actuated joint and the positions of two hands and two feet of the humanoid. At each time step, we randomly sample the state transitions from the demonstrations and feed them into the discriminator to get the expert prediction loss for enabling the discriminator to distinguish them,

$$\mathcal{L}_{expert} = \mathbb{E}_{(o_t^D, o_{t+1}^D)\sim\mathcal{D}}[(D(o_t^D, o_{t+1}^D) - 1)^2] \tag{2}$$

As do the same with state transitions sampled from the policy,

$$\mathcal{L}_{policy} = \mathbb{E}_{(o_t^D, o_{t+1}^D)\sim\pi}[(D(o_t^D, o_{t+1}^D) + 1)^2] \tag{3}$$

We follow to [27] to formulate the penalty for gradients on samples from the reference to stabilize training,

$$\mathcal{L}_{GP} = \mathbb{E}_{(o_t^D, o_{t+1}^D)\sim\mathcal{D}} \left[\|\nabla \mathcal{D}(o_t^D, o_{t+1}^D)\|^2 \right] \tag{4}$$

Finally, we formulate the total AMP loss as,

$$\mathcal{L}_{AMP} = \frac{1}{2}\mathcal{L}_{expert} + \frac{1}{2}L_{policy} + \lambda_{GP}\mathcal{L}_{GP} \tag{5}$$

The AMP loss function directs the discriminator to rate samples, giving scores close to +1 for genuine reference motions and nearing -1 for those generated by the policy.

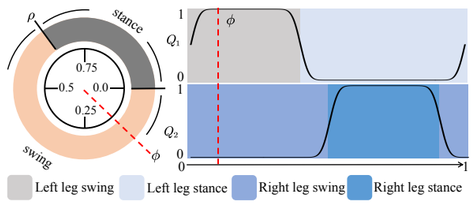


Fig. 4: Visualization of periodic rewards based on the von Mises distribution

The objective of the policy is to create motions convincing enough to lead the discriminator into assigning higher scores, demonstrating its ability to closely mimic reference motions. Subsequently, the formulation of the imitation reward for the policy's training is denoted as,

$$r_I = \max[0, 1 - \frac{1}{4}(D(o_t^D, o_{t+1}^D) - 1)^2] \tag{6}$$

where o_t^D, o_{t+1}^D are sampled from the policy.

B. End-to-End Reinforcement Learning on Humanoid Robot

Concurrently, the movement direction in reference motion is typically limited to the local coordinate. To facilitate control under the world coordinate system, generate more natural gaits, and enable a more effective transition from simulation to reality on challenging terrains, we introduced coordinated task rewards. The task rewards are composed of three parts, the command reward, the periodic reward, and the regularization reward. The command reward forces the robot to move alone in the command directions, which is formulated as,

$$r_{com} = \sum \lambda_i \exp(-\omega_i(|\mathbf{v_{des}^i} - \mathbf{v_t^i}|) \quad i \in (x, y, yaw) \tag{7}$$

where λ_i and ω_i are the weight of each direction of command rewards, $\mathbf{v}_{des}$ is the desire velocity vector along the specific direction and $\mathbf{v}_t$ is the velocity vector at t.

To facilitate the attainment of the desired gait performance, we introduce periodic rewards aligned with imitation rewards. This method naturally promotes the maintenance of a stable gait in the robot. However, if a variable gait is desired, it would be advisable to omit this reward function. We follow [9] to formulate the periodic reward by the swing phases, characterized by the foot moving through the air, and the stance phases, where the foot is firmly planted on the ground. Each periodic reward item is composed of a coefficient α_i, a phase indicator $I_i(\phi)$, a valued phase reward function $V_i(s_t)$, ϕ is the cycle time, i denotes the phase is stance phase or swing phase. The swing and stance phases are sequentially arranged and collectively span the entire cycle duration by establishing a ratio $\rho \in (0, 1)$. This configuration ensures that the swing phase occupies a duration equivalent to ρ, followed immediately by the stance phase which extends for a duration of $1 - \rho$. The single-foot reward is written as follows,

$$\begin{aligned} r_{per} &= \sum \alpha_i \mathbb{E}[I_i(\phi)] V_i(s_t) \\ V_{stance}(s_t) &= \exp(-10F_f^2) \\ V_{swing}(s_t) &= \exp(-200v_f^2) \end{aligned} \tag{8}$$

environment
cross-validation
real robot
deploy
feedback
r_{task}
o_t
a_t
ϕ
c_t
policy
r_I
$o_t^D \sim \pi$
discriminator
$o_t^D \sim \mathcal{D}$
re-targeting
reference motion

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ρ
stance
swing
0.75
0.5
0.0
0.25
ϕ
1
Q_1
0
Q_2
Left leg swing
Left leg stance
Right leg swing
Right leg stance

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$$\begin{aligned} Q_1 &= I_{stance}(\phi + \theta_{left}) \\ Q_2 &= I_{stance}(\phi + \theta_{right}) \end{aligned} \tag{9}$$

where the θ_{left}, θ_{right} is the offsets of left and right leg in cycle time. Expect the normal periodic reward, we calculate rewards for foot speed, height difference, and the symmetric during the swing phase for more natural gait style. The foot speed tracking reward is formulated as,

$$\begin{aligned} q^i &= \text{clip}(\frac{\phi}{\rho} - 0.5, 0, 1) \\ r(s_t) &= \begin{cases} 16(q^i v_f^i)^2 & 0 \leq q_i \leq 0.6, \\ 0 & q_i > 0.6. \end{cases} \end{aligned} \tag{10}$$

where $i \in (\text{left}, \text{right})$, "clip" is the function to constrain the values of variables within the range of 0 to 1. The foot speed tracking reward encourages the robots to perform higher foot speed during the swing phase. The height difference reward is,

$$\begin{aligned} q^i &= \frac{\phi}{\rho} \\ \delta h &= h_f^i - h_f^{-i} - 0.02 \\ r(s_t) &= \begin{cases} 2\text{exp}(-25(|\delta h|) & 0 \leq q_i \leq 0.3, \\ 0 & q_i > 0.3. \end{cases} \end{aligned} \tag{11}$$

where h_f^i is the height i foot, the h_f^{-i} denotes the height of the other side foot. The purpose of this function is to calculate reward based on the height difference of the foot only during certain early stages of the gait cycle. The symmetric reward is written as follows,

$$\begin{aligned} \mathbf{d}_t &= \mathbf{p}_t^{left} - \mathbf{p}_t^{right} \\ tf &= (\mathbb{E}[I_{left}(\phi)] > 0.5) \wedge (\mathbb{E}[I_{right}(\phi)] > 0.5) \\ \delta \mathbf{f}_t &= tf \cdot \mathbf{d}_t + \neg tf \cdot \delta \mathbf{f}_{t-1} \\ \delta \mathbf{l}_t &= \neg tf \cdot \delta \mathbf{f}_t + tf \cdot \mathbf{d}_t \\ r(s_t) &= 3.3tf\text{exp}(-10||\mathbf{d}_t + \delta \mathbf{l}_t||_1) \end{aligned} \tag{12}$$

where $\mathbf{p}_t^i$ is the 3-D position of foot end-effector, $\neg$ is negation symbol. This function calculates a reward based on the symmetry of the gait swing, where the gait phases of the left and right feet are taken into consideration, along with the distance between the feet.

To enhance the robustness of the sim-to-real transfer, we incorporated regularization rewards into the comprehensive reward structure. These rewards enforce motion constraints, emphasizing both smoothness and safety. The detailed formulation of each reward is shown in Table II. The DoF limits reward and keeps the movement within the robot's physical capabilities, ensuring that it doesn't attempt movements that could damage its mechanisms or be outside its operational scope. $\mathbf{b}$ denotes the vector angle of DoF. $\mathbf{b}_{lower}$ and $\mathbf{b}_{upper}$ are the upper and lower bounds of the joint limit. The DoF velocity reward forces the system to maintain optimal speeds. Similarly, the DoF acceleration reward encourages smooth

TABLE II: Regularization Rewards

Item	Detail
Action differential	$\text{exp}(-0.05\|\mathbf{a}_t - \mathbf{a}_{t-1}\|_2)$
DoF limits	$\text{exp}(-2.0(\text{min}(0, \mathbf{b} - \mathbf{b}_{upper})$ $-\text{max}(0, \mathbf{b} - \mathbf{b}_{lower})))$
DoF velocity	$\text{exp}(-1 \times 10^{-4}\|\dot{\mathbf{b}}\|_2^2)$
DoF acceleration	$\text{exp}(-1 \times 10^{-7}\|\ddot{\mathbf{b}}\|_2^2)$
Arm DoF penalty	$\text{exp}(\|\mathbf{b}_{arm}\|_1)$
Orientation differential	$\text{exp}(-300(roll^2 + pitch^2))$
Torso yaw	$\|b_{torso\ yaw}\|_1$
Torques	$\text{exp}(-5 \times 10^{-4}\|\tau_t\|_2)$

TABLE III: Domain Randomization Range

Randomization Item	Range	Unit
Mass	[-0.05,0.05]	Kg
Center of mass alone x	[-0.05,0.05]	m
Center of mass alone z	[-0.05,0.05]	m
Motor strength	[0.7,1.4]×default	N·m
Impulse	[0,0.8]	m/s
External force	[-500,500]	N
Linear velocity	[0.8,1.2]×default	m/s

increases and decreases in speed, contributing to the overall stability. For robotic arms, the arm DoF penalty is applied to discourage large positions or movements. The torso yaw reward and orientation differential rewards are specific to the rotational movement around a robot's vertical axis. This control is vital for balance and orientation, especially in humanoid robots.

V. Experiments

A. Training and Implementation Details

1) Training Setup: We trained our policy via a model-free reinforcement learning algorithm known as Proximal Policy Optimization (PPO) [37] on 4096 Isaac Gym simulation environments parallel. In humanoid locomotion scenarios, actions are represented by $a_t \in \mathbb{R}^{25}$ a 25-dimensional vector specifying the desired positional adjustments for each actuated joint as dictated by the Proportional-Derivative (PD) controller. Observations, denoted as $o_t \in \mathbb{R}^{91}$, encompass the humanoid's current linear and angular velocities, the average velocities in the (x, y, yaw) directions, and the orientation of the gravity vector within the robot's base frame, all measured by an inertial measurement unit (IMU). The position and velocity of each joint are captured by the actuators' encoders. A command, $c_t = (v_{des}^x, v_{des}^y, \omega_{des}^{yaw})$, is introduced to specify the desired velocities along the x-axis, y-axis, and yaw within the robot's base frame. Periodic observations include the sine and cosine of the cycle time and the swing phase ratio ρ, to dynamically adjust the gait. Additionally, the root's height is incorporated into the observations to manage the humanoid robot's posture. For enhanced performance, the action executed in the previous timestep is also considered. Our framework ensures a smooth training process for Adam, as illustrated in Figure 6.

2) Domian Randomization: This randomization strategy aims to tackle three critical uncertainties: modeling discrepancies in the robot and its environment, sensor noise, and unobservable disturbances and impulses. To avoid the sim-to-real gap in force applying in Isaac Gym, we use the sudden velocity on the root to replace the force impulse. These dynamic properties are represented in Table III, with

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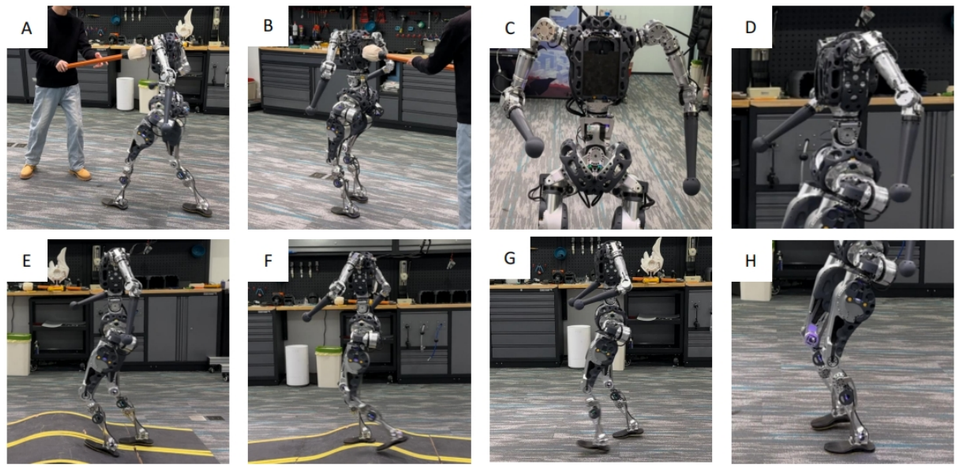


Fig. 5: **Real Robot Experiments.** We tested our method on Adam. (A)(B) demonstrate the robot's robust locomotion performance under external disturbances. **Especially, it demonstrated the characteristic of "straight-knee"** (C)(D) show the human-like swinging of the upper limbs. (E)(F) display the robustness and human-likeness on unknown and complex terrains. **(G)(H) present for the first time the humanoid robot's "heel-to-toe" running and walking gaits.**

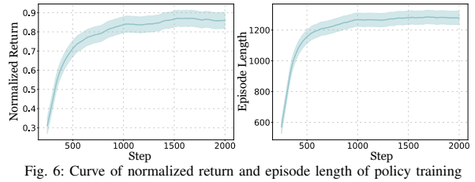


Fig. 6: Curve of normalized return and episode length of policy training

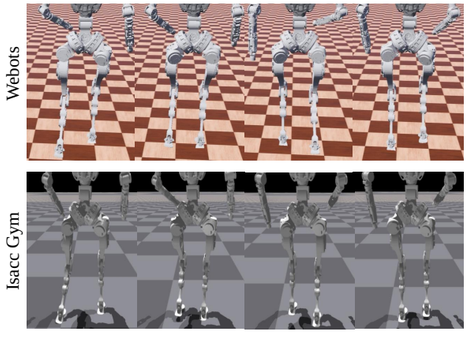


Fig. 7: Cross-validation of Webots and Isaac gym simulator in forward locomotion scenario

values adjusted across the specified ranges. The mass of individual robot bodies (*e.g.*, torso), is varied to simulate different weight distributions. The center of mass is randomly shifted in specific directions. These techniques help the policy to adapt to the uncertainties in robot components. Motor strength influences the torque output to deal with the uncertainties in motors. Furthermore, impulses are applied by adding velocities in the x-y plane on the robot root, replicating the effect of sudden external shocks. External forces are also directly applied to the body of the robot to simulate unobservable environmental forces. Additionally, noise is injected into the linear velocity to simulate speed estimation errors. A series of tests were performed under a variety of conditions and in response to external disturbances, with the outcomes illustrated in Figure. 5. These results shows our method provided a new perspective of future motion learning for humanoid robots

B. Cross-validation and Feedback Fine-tuning

We carried out comprehensive cross-validation experiments using both Webots [10] and Isaac Gym as simulation platforms to evaluate the effectiveness of our models in environments that closely mimic real-world conditions. Notably, Webots offers advanced physical simulation capabilities, enabling us to validate our policy across diverse platforms while fine-tuning hyperparameters for optimal performance. Which vividly demonstrated in Figure 7, where the humanoid robot's gait appears synchronized and uniform across both simulators, underscoring the model's reliability. Additionally, we conducted real-world tests to further validate our model's practical applicability. The seamless transition of the humanoid robot from a standing to a running state in the real world is captured in Figure 1, highlighting the model's effectiveness and the potential for real-world deployment.

VI. Conclusion

In this paper, we implement a whole-body imitation learning framework on the humanoid robot Adam, whose performance in complex gaits tasks is comparable to that of humans, and demonstrate human-like features such as "heel-to-toe" transitions and straight-knee movements for the first time in humanoid robots. Our experiments have demonstrated the practicality and efficiency of our framework, highlighting its great potential for further research in humanoid robotics. Moreover, our framework supports the integration of perceptual modules, and in the future, we plan to incorporate more sensors to allow Adam to imitate human motions under integrated perceptual conditions.

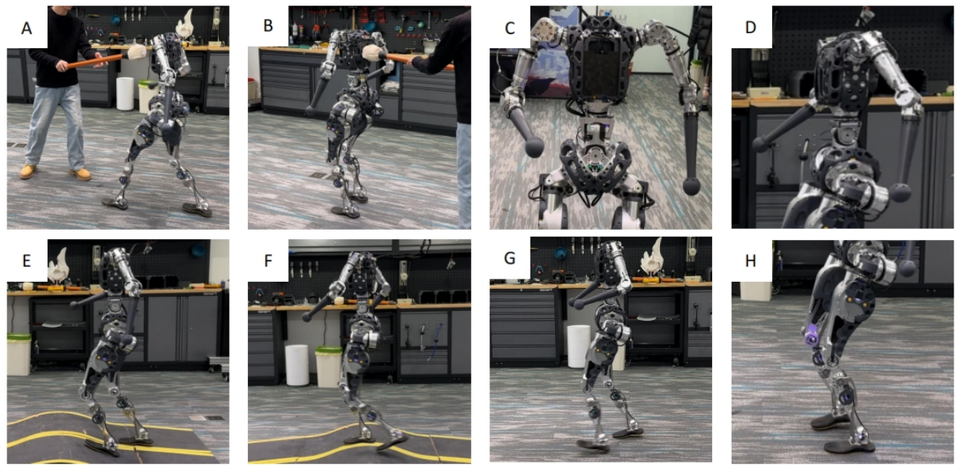


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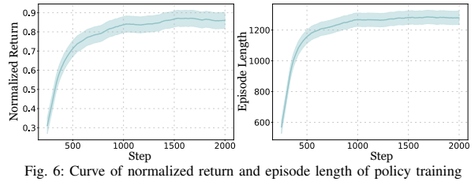


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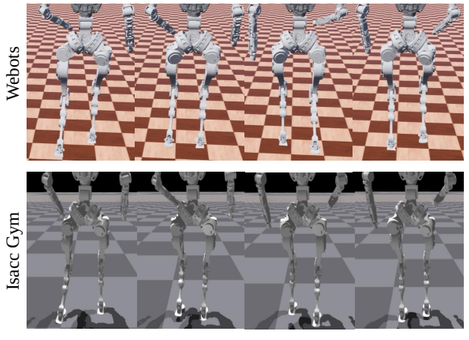


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References

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[4] Z. Li, X. Cheng, X. B. Peng, P. Abbeel, S. Levine, G. Berseth, and K. Sreenath, "Reinforcement learning for robust parameterized locomotion control of bipedal robots," in *IEEE International Conference on Robotics and Automation (ICRA)*, Xi'an, China, June 2021.

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[6] N. Rudin, D. Hoeller, P. Reist, and M. Hutter, "Learning to walk in minutes using massively parallel deep reinforcement learning," in *Conference on Robot Learning*. PMLR, 2022, pp. 91–100.

[7] Z. Li, X. B. Peng, P. Abbeel, S. Levine, G. Berseth, and K. Sreenath, "Robust and versatile bipedal jumping control through reinforcement learning."

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[10] D. Crowley, J. Dao, H. Duan, K. Green, J. Hurst, and A. Fern, "Optimizing bipedal locomotion for the 100m dash with comparison to human running," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 12 205–12 211.

[11] S. H. Jeon, S. Heim, C. Khazoom, and S. Kim, "Benchmarking potential based rewards for learning humanoid locomotion," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 9204–9210.

[12] F. Shi, Y. Kojio, T. Makabe, T. Anzai, K. Kojima, K. Okada, and M. Inaba, "Reference-free learning bipedal motor skills via assistive force curricula," in *The International Symposium of Robotics Research*. Springer, 2022, pp. 304–320.

[13] R. P. Singh, Z. Xie, P. Gergondet, and F. Kanehiro, "Learning bipedal walking for humanoids with current feedback," *IEEE Access*, 2023.

[14] D. Kim, G. Berseth, M. Schwartz, and J. Park, "Torque-based deep reinforcement learning for task-and-robot agnostic learning on bipedal robots using sim-to-real transfer," *arXiv preprint arXiv:2304.09434*, 2023.

[15] T. Haarnoja, B. Moran, G. Lever, S. H. Huang, D. Tirumala, M. Wulfmeier, J. Humplik, S. Tunyasuvunakool, N. Y. Siegel, R. Hafner, *et al.*, "Learning agile soccer skills for a bipedal robot with deep reinforcement learning," *arXiv preprint arXiv:2304.13653*, 2023.

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[17] A. Tang, T. Hiraoka, N. Hiraoka, F. Shi, K. Kawaharazuka, K. Kojima, K. Okada, and M. Inaba, "Humanmimic: Learning natural locomotion and transitions for humanoid robot via wasserstein adversarial imitation," *arXiv preprint arXiv:2309.14225*, 2023.

[18] T. Osa, J. Pajarinen, G. Neumann, J. A. Bagnell, P. Abbeel, J. Peters, *et al.*, "An algorithmic perspective on imitation learning," *Foundations and Trends® in Robotics*, vol. 7, no. 1-2, pp. 1–179, 2018.

[19] H. Ravichandar, A. S. Polydoros, S. Chernova, and A. Billard, "Recent advances in robot learning from demonstration," *Annual review of control, robotics, and autonomous systems*, vol. 3, pp. 297–330, 2020.

[20] S. Bohez, S. Tunyasuvunakool, P. Brakel, F. Sadeghi, L. Hasenclever, Y. Tassa, E. Parisotto, J. Humplik, T. Haarnoja, R. Hafner, *et al.*, "Imitate and repurpose: Learning reusable robot movement skills from human and animal behaviors," *arXiv preprint arXiv:2203.17138*, 2022.

[21] L. Han, Q. Zhu, J. Sheng, C. Zhang, T. Li, Y. Zhang, H. Zhang, Y. Liu, C. Zhou, R. Zhao, *et al.*, "Lifelike agility and play on quadrupedal robots using reinforcement learning and generative pre-trained models," *arXiv preprint arXiv:2308.15143*, 2023.

[22] S. Schaal, "Is imitation learning the route to humanoid robots?" *Trends in cognitive sciences*, vol. 3, no. 6, pp. 233–242, 1999.

[23] J. Van Den Berg, S. Miller, D. Duckworth, H. Hu, A. Wan, X.-Y. Fu, K. Goldberg, and P. Abbeel, "Superhuman performance of surgical tasks by robots using iterative learning from human-guided demonstrations," in *2010 IEEE International Conference on Robotics and Automation*. IEEE, 2010, pp. 2074–2081.

[24] P. M. Kebria, A. Khosravi, S. M. Salaken, and S. Nahavandi, "Deep imitation learning for autonomous vehicles based on convolutional neural networks," *IEEE/CAA Journal of Automatica Sinica*, vol. 7, no. 1, pp. 82–95, 2020.

[25] L. Le Mero, D. Yi, M. Dianati, and A. Mouzakitis, "A survey on imitation learning techniques for end-to-end autonomous vehicles," *IEEE Transactions on Intelligent Transportation Systems*, 2022.

[26] J. Ho and S. Ermon, "Generative adversarial imitation learning," *Advances in neural information processing systems*, vol. 29, 2016.

[27] X. B. Peng, Z. Ma, P. Abbeel, S. Levine, and A. Kanazawa, "Amp: Adversarial motion priors for stylized physics-based character control," *ACM Transactions on Graphics (ToG)*, vol. 40, no. 4, pp. 1–20, 2021.

[28] X. B. Peng, Y. Guo, L. Halper, S. Levine, and S. Fidler, "Ase: Large-scale reusable adversarial skill embeddings for physically simulated characters," *ACM Transactions On Graphics (TOG)*, vol. 41, no. 4, pp. 1–17, 2022.

[29] A. Escontrela, X. B. Peng, W. Yu, T. Zhang, A. Iscen, K. Goldberg, and P. Abbeel, "Adversarial motion priors make good substitutes for complex reward functions," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2022, pp. 25–32.

[30] E. Vollenweider, M. Bjelonic, V. Klemm, N. Rudin, J. Lee, and M. Hutter, "Advanced skills through multiple adversarial motion priors in reinforcement learning," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 5120–5126.

[31] C. Li, M. Vlastelica, S. Blaes, J. Frey, F. Grimminger, and G. Martius, "Learning agile skills via adversarial imitation of rough partial demonstrations," in *Conference on Robot Learning*. PMLR, 2023, pp. 342–352.

[32] J. Wu, G. Xin, C. Qi, and Y. Xue, "Learning robust and agile legged locomotion using adversarial motion priors," *IEEE Robotics and Automation Letters*, 2023.

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[34] K. Ayusawa and E. Yoshida, "Motion retargeting for humanoid robots based on simultaneous morphing parameter identification and motion optimization," *IEEE Transactions on Robotics*, vol. 33, no. 6, pp. 1343–1357, 2017.

[35] R. Grandia, F. Farshidian, E. Knoop, C. Schumacher, M. Hutter, and M. Bächer, "Doc: Differentiable optimal control for retargeting motions onto legged robots," *ACM Transactions on Graphics (TOG)*, vol. 42, no. 4, pp. 1–14, 2023.

[36] J. H. Park, "Impedance control for biped robot locomotion," *IEEE Transactions on Robotics and Automation*, vol. 17, no. 6, pp. 870–882, 2001.

[37] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

Notes

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[6] https://openai.com/
[7] https://www.1x.tech/
[8] https://mocap.cs.sfu.ca/
[9] http://mocap.cs.cmu.edu/
[10] https://cyberbotics.com/

References

[1] Wikipedia, "Optimus (robot)," https://en.wikipedia.org/wiki/Optimus_(robot), 2024, accessed: 2024-02-28.

[2] Y. Gong, R. Hartley, X. Da, A. Hereid, O. Harib, J.-K. Huang, and J. Grizzle, "Feedback control of a cassie bipedal robot: Walking, standing, and riding a segway," in *2019 American Control Conference (ACC)*. IEEE, 2019, pp. 4559–4566.

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[8] https://mocap.cs.sfu.ca/
[9] http://mocap.cs.cmu.edu/
[10] https://cyberbotics.com/